

# Ahmed Moussaoui

541-908-6657 | [ahmedamoussaoui@gmail.com](mailto:ahmedamoussaoui@gmail.com) | [ahmedmou.github.io](https://ahmedmou.github.io) | Corvallis, OR (Open to Relocation)  
[linkedin.com/in/ahmed-latif-moussaoui](https://linkedin.com/in/ahmed-latif-moussaoui)

## EDUCATION

### Oregon State University

*B.S. in Mechanical Engineering, Robotics Control Option*

Corvallis, OR

*Expected June 2026*

## WORK EXPERIENCE

### Robotics Lab Engineer

Feb. 2025 – Present

*OSU College of Civil Engineering*

*Corvallis, OR*

- Led a 2-3 person team designing and deploying multiple automation systems for a civil engineering research lab, targeting repetitive testing processes with a 7-DOF robotic manipulator.
- Built a vision-guided pick-and-place pipeline from scratch, reducing manual technician task time by approximately 75% through automated object detection, grasp pose generation, and motion planning.
- Presented system capabilities and progress to a group of potential investors on behalf of the research group.
- Containerized the completed system using Docker for reproducible deployment across lab environments.

### Undergraduate Learning Assistant, ME 203

Sept. 2025 – Present

*OSU MIME Department*

*Corvallis, OR*

- Supported 30+ students per term in Python programming for engineering data analysis, numerical methods, and visualization.
- Led lab sessions and provided one-on-one guidance, collaborating with faculty to refine course materials.

### 3D Modeler

May 2022 – June 2022

*Freelancer, Fiverr*

*Remote*

- Produced manufacturing-ready CAD models for multiple clients across 3D printing, injection molding, and CNC machining. Maintained a 5-star rating across all orders.

## PROJECTS

### Autonomous RC Aircraft | ArduPilot, Mission Planner, Avionics, Fabrication

Sept. 2024 – June 2025

- Autonomous sub-team lead on a 6-10 person DBF capstone team. Built the aircraft from scratch including frame, full wiring harness, and avionics integration.
- Achieved multiple successful autonomous test flights across several test days, the first in the team's history.

### SAMI Humanoid Robot – Face Display and Head Design | ROS2, Pygame, Fusion 360, Raspberry Pi

Spring 2025

- One of 3-5 person team. Designed and fabricated the full robot head enclosure and screen mount in Fusion 360. Built an animated face display with real-time ROS2 emotion state control.
- Delivered a successful live demo with continuous reliable operation in front of an audience.

### Automated Security Door System | C++, Arduino, Fusion 360, 3D Printing

OSU ME 351

- Designed and 3D printed the full mechanical enclosure, integrating servo actuation, linkages, and sensors.
- Built a custom metal detector circuit from scratch and wrote a 5-state C++ control system with jam detection, forced entry detection, and manual override.

### Lunar Mining Rover Chassis | CAD, 3D Printing, Mechanical Design

Sept. 2020 – June 2021

- Designed rover frame and motor mounts in CAD for the LBCC Space Exploration Club, iterated on fit and function through 3D printed prototypes.

## TECHNICAL SKILLS

**CAD and Design:** SolidWorks, Fusion 360, Autodesk Inventor, GD&T, DFM, Part and Assembly Design

**Fabrication:** 3D Printing (FDM), Rapid Prototyping, Machining, Soldering, Wiring and Electrical Assembly, Machine Shop

**Robotics:** ROS2, MoveIt2, ArduPilot, Mission Planner, Motion Planning, Systems Integration, Embedded Systems

**Programming:** Python, C++, MATLAB, Arduino, OpenCV, YOLO, Docker, Linux, GitHub

**AI and ML:** AI API Integration, Computer Vision (OpenCV, YOLO), Pre-trained Model Deployment, Model Fine-tuning

**Platforms:** Linux, Raspberry Pi, Raspberry Pi OS

## LEADERSHIP

### Head Coach, Esports Club

Sept. 2024 – Present

*Oregon State University*

*Corvallis, OR*

- Coached and managed approximately 20 players across two competitive Valorant teams, improving on-field coordination by 50% through targeted communication and teamwork drills.

### Vice President, LEGO Club

Nov. 2024 – Present

*Oregon State University*

*Corvallis, OR*

- Co-founded the club with faculty support, grew membership to 20+ students in the first term through weekly build meetings and hands-on engineering activities.